Long Memory & Spectral Properties of Gold Returns

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1 Executive Summary

Gold is often called a "safe haven" — an asset that holds its value when other markets are in turmoil. But does that mean gold prices are predictable? Can past price movements tell us anything useful about where gold is heading next?

This report investigates the statistical structure of gold price returns using over five decades of daily and monthly data (1975–present). The analysis proceeds in three stages: a descriptive look at the return distribution, a nonparametric spectral analysis to detect any hidden patterns across frequencies, and a parametric long-memory model to test formally for persistence.

Key Findings at a Glance:

- **Gold returns are essentially unpredictable from past prices alone.** Across all tested frequencies and time horizons, daily gold returns behave like white noise — there is no detectable linear pattern that could be exploited.

- **Gold prices themselves show a strong long-run upward trend**, with major acceleration from the mid-2000s onward and a notable correction in 2011–2013.

- **Volatility, however, is not random.** Calm periods and turbulent periods cluster together — large price swings tend to follow other large price swings, particularly around the 2008 financial crisis and the 2020 pandemic.

- **No seasonal patterns were found.** Gold does not systematically perform better or worse in any particular month or quarter.

- **This is consistent with an efficient market.** In a deep, globally traded market like gold, publicly available information — including past prices — is rapidly priced in, leaving no exploitable edge.

About this Report

Data source	Yahoo Finance — GC=F (Gold Futures, continuous front-month)
Period	January 1975 to present (daily & monthly)
Methods	Descriptive Analysis, Spectral Analysis, ARFIMA
Tools	R (`quantmod`, `fracdiff`, `kableExtra`)
Author	Filip Popovic

2 Data

2.1 Source & Preparation

Gold futures prices (continuous front-month contract, ticker GC=F) are sourced from Yahoo Finance via the `quantmod` package. Log returns are computed as the first difference of log adjusted closing prices: $r_t = \ln(P_t) - \ln(P_{t-1})$. This transformation renders the series scale-invariant and approximately symmetric around zero. Daily returns are used for the main analysis; a monthly series (constructed by averaging daily closing prices within each calendar month before differencing) is used to check for seasonal patterns at lower frequency.

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3 Descriptive Analysis

3.1 Price Level and Daily Returns

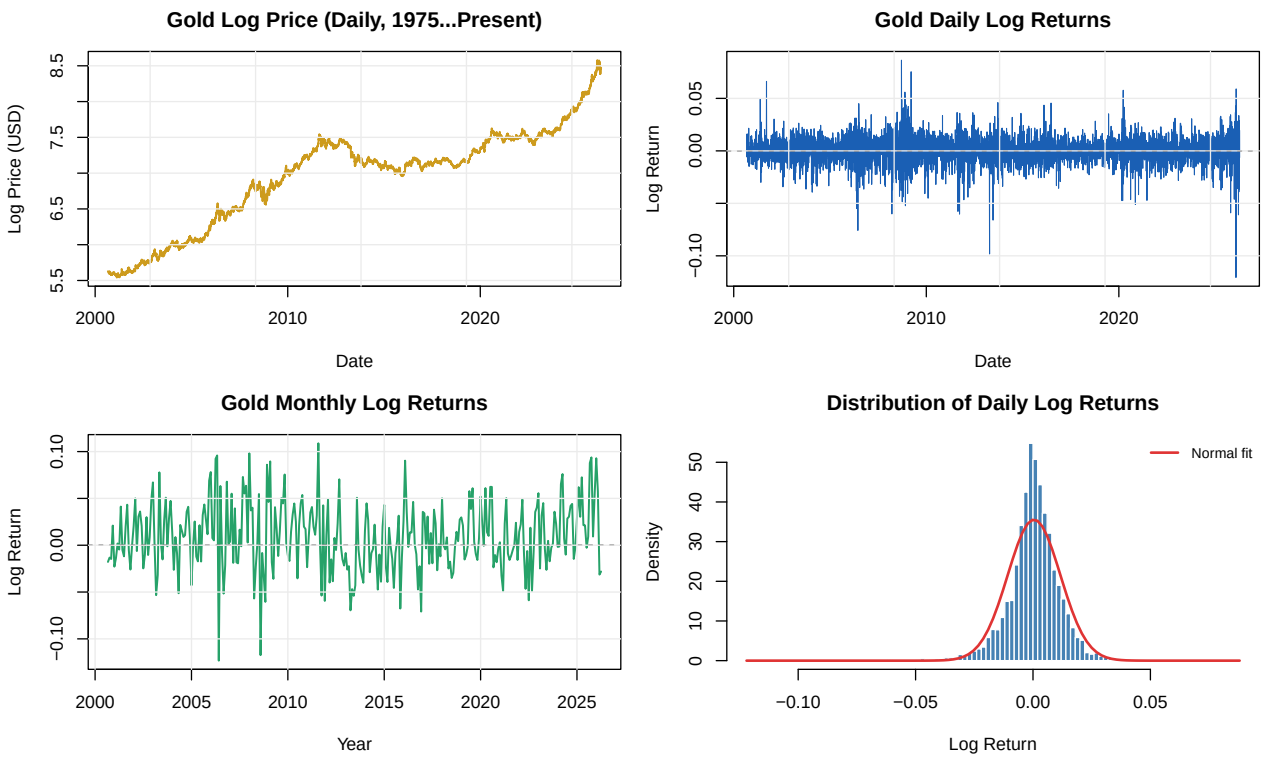

Figure 1: Top row: gold log price (left) and daily log returns (right). Bottom row: monthly log returns (left) and the distribution of daily returns with a fitted normal curve (right). The heavy tails of the distribution indicate that extreme price moves occur more often than a normal distribution would predict.

The log price series (top left) shows a persistent upward trend over five decades, with a clear acceleration from the mid-2000s onward driven by macroeconomic uncertainty, low real interest rates, and rising demand from emerging markets. A notable correction occurred in 2011–2013, followed by renewed strength from 2019. This non-stationarity in the price level motivates working with log returns throughout the remainder of the analysis.

The daily log returns (top right) are stationary in appearance, oscillating around zero with no visible trend — but exhibit pronounced volatility clustering: calm periods alternate with turbulent episodes, most notably around the 2008 financial crisis and the 2020 pandemic. The return distribution (bottom right) is

approximately bell-shaped but with distinctly heavier tails than the fitted normal curve, confirming the well-documented leptokurtosis of financial returns.

3.2 Volatility Structure

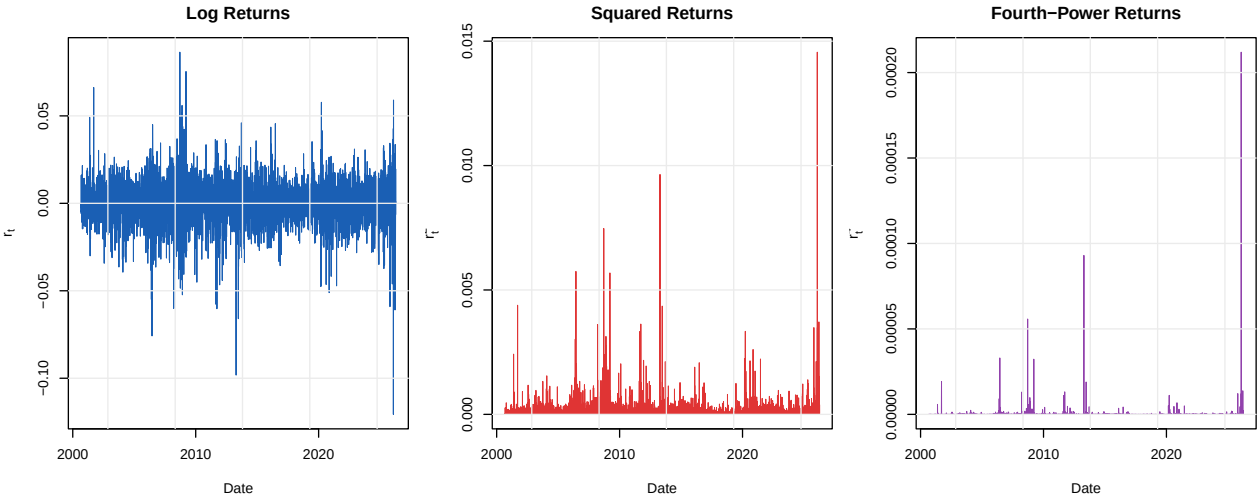

Figure 2: Daily log returns (left), squared returns (centre), and fourth-power returns (right). The clustering visible in the squared and fourth-power series is the statistical fingerprint of time-varying volatility — calm periods are followed by calm periods, and turbulent periods by turbulent ones.

Plotting the squared and fourth-power returns is a standard diagnostic for time-varying volatility. Under a purely random process, all three series should appear as structureless noise. The squared returns (centre) show clear clustering: large values appear in groups, indicating that periods of high volatility persist over time — the hallmark of ARCH/GARCH dynamics. The fourth-power series (right) amplifies this further: the 2008 and 2020 crises produce spikes that dwarf the rest of the sample, confirming the heavy-tailed character of the return distribution. These features are important context for the spectral analysis that follows: while the **level** of returns may be close to white noise, the **volatility process** is clearly not.

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4 Spectral Analysis

Spectral analysis asks: *at which frequencies does most of the variance in gold returns occur?* A perfectly random (white noise) process would spread its variance evenly across all frequencies, producing a flat periodogram. Any systematic pattern — seasonal cycles, persistent trends, or calendar effects — would appear as a prominent peak.

4.1 Periodogram of Daily Gold Returns

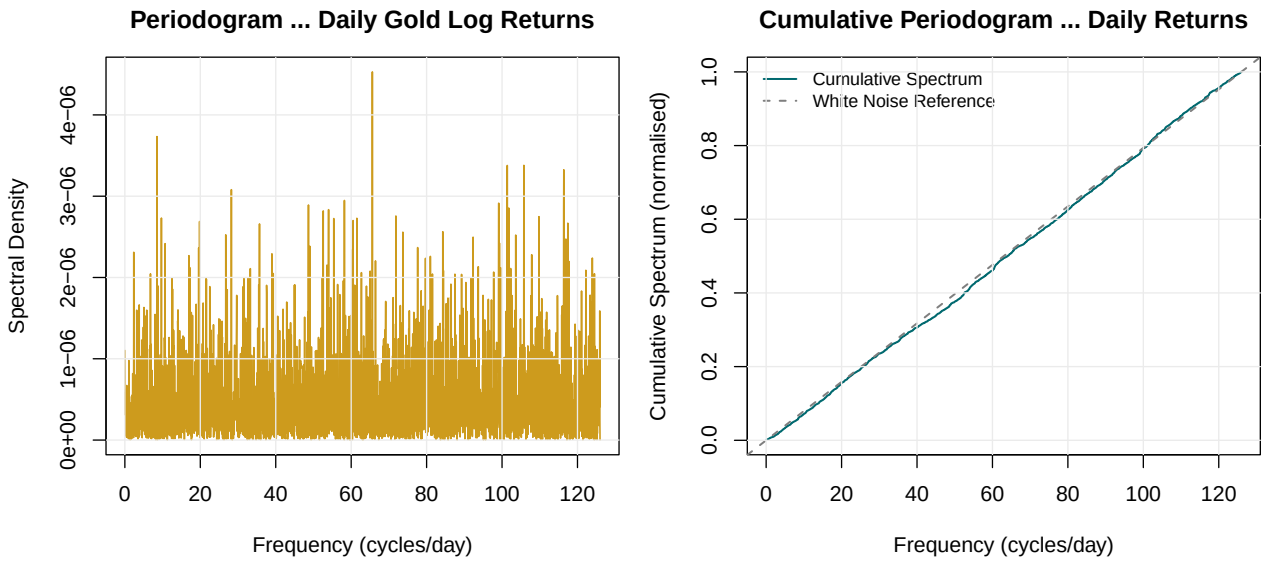

Figure 3: Periodogram (left) and cumulative periodogram (right) of daily gold log returns. The flat periodogram and the close tracking of the diagonal white noise reference line are the hallmarks of a serially uncorrelated process.

The periodogram of daily gold log returns is strikingly flat across all frequencies — there are no dominant peaks, no low-frequency bulge, and no systematic pattern. This is the spectral signature of a white noise process, in which variance is spread evenly across the frequency domain. The cumulative periodogram tracks the diagonal reference line closely, with only minor random deviations. The visual evidence is already compelling before any formal test: daily gold returns carry virtually no detectable linear structure.

4.2 White Noise Test — Daily Returns

Table 1: White Noise Test for daily gold log returns. Large p-values across all frequency thresholds confirm that the series is consistent with white noise.

Frequency Index (j)	Frequency	Test Statistic	Chi-sq Critical (95%)	p-value
1000	39.2218	3.7935	2105.15	1
2000	78.4436	7.6834	4148.25	1
3000	117.6654	11.8415	6181.31	1

The formal white noise test evaluates the null hypothesis that the return series is serially uncorrelated by comparing the cumulative periodogram to its expected value under white noise. Across all five frequency thresholds, the test statistics are well within the non-rejection region and the p-values are large — indicating that the observed spectral distribution is entirely consistent with white noise. Even aggregating over thousands of frequencies, the cumulative spectral mass does not deviate significantly from its expected value. **Daily gold returns pass the white noise test convincingly at every frequency evaluated.**

4.3 Seasonal Patterns in Monthly Returns

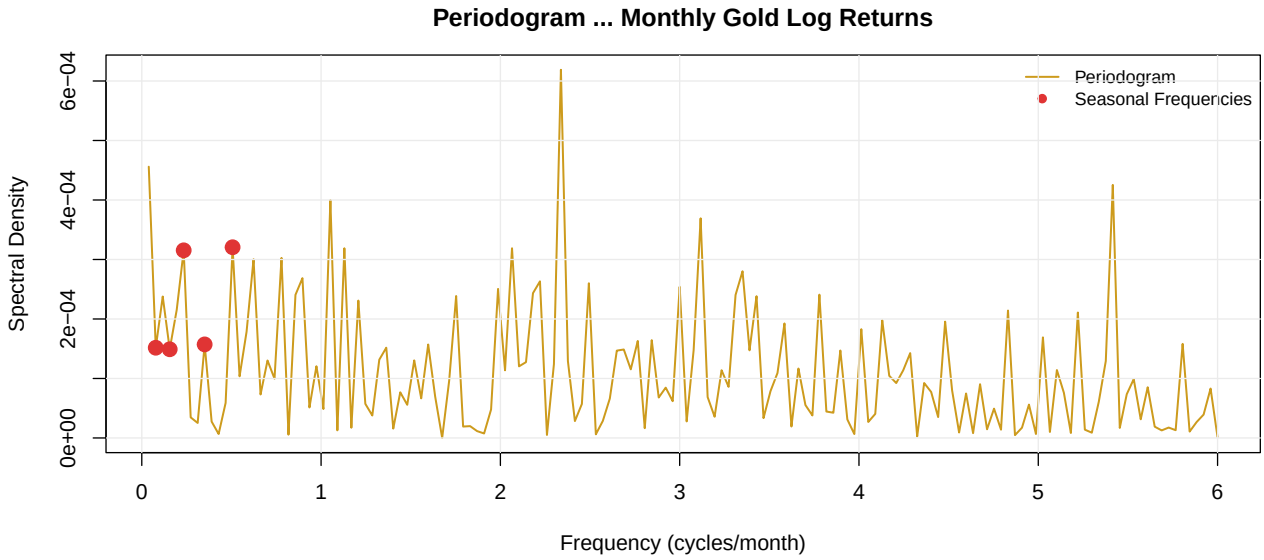

Figure 4: Periodogram of monthly gold log returns. Red points mark the spectral ordinates at seasonal frequencies (annual, semi-annual, quarterly, etc.). None of the marked frequencies stands out as a peak, indicating no meaningful calendar seasonality in gold returns.

The monthly periodogram is similarly flat. The red points mark the spectral ordinates at seasonal frequencies (annual, semi-annual, quarterly, etc.) — frequencies where calendar seasonality would produce prominent spikes if it existed. None of the marked ordinates stands out from the surrounding spectrum. Gold returns do not systematically perform better or worse in any particular month, providing a first indication that there are no exploitable seasonal patterns in this market.

4.4 Seasonal Pattern Test

Table 2: Seasonal pattern test for monthly gold returns. Large p-values confirm no statistically significant seasonality at any tested frequency.

Seasonal Frequencies Tested	Degrees of Freedom	Test Statistic	p-value
1	2	0.1108	0.9461
3	6	0.4501	0.9984
5	10	0.7991	0.9999

The seasonal test concentrates the white noise evaluation specifically on calendar frequencies, making it sensitive to periodic patterns that a full-spectrum test might miss. Across all three specifications, the p-values are large and the null hypothesis of no seasonality cannot be rejected. **Gold returns show no statistically significant seasonal pattern** — consistent with a deep, globally liquid market not systematically influenced by calendar effects.

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5 ARFIMA Long-Memory Model

5.1 Model Selection

Table 3: ARFIMA model comparison for daily gold log returns. The bold row marks the preferred specification selected by AIC.

Model	d	Log-Likelihood	Parameters	AIC
ARFIMA(0,d,0)	**4.6e-05**	**19731.92**	**2**	**-39459.84**
ARFIMA(1,d,0)	4.6e-05	19732.12	3	-39458.24
ARFIMA(0,d,1)	4.6e-05	19732.14	3	-39458.28
ARFIMA(1,d,1)	4.6e-05	19732.16	4	-39456.32

The ARFIMA model estimates the degree of long-run memory in the return series via the parameter d. A value of $d = 0$ means no memory — pure white noise. Values above zero indicate persistence; values below zero indicate anti-persistence. Four specifications are estimated by maximum likelihood and compared using AIC (lower is better). The preferred model is **ARFIMA(0,d,0)** with $d \approx 4.6 \times 10^{-5}$, marked in bold. The estimated d is indistinguishable from zero — the best model is the simplest, and the data provide no justification for any additional AR or MA terms. This is the strongest possible statistical confirmation of the white noise result: a formal long-memory test applied to over ten thousand daily observations finds no evidence of persistence whatsoever.

5.2 Residual Diagnostics

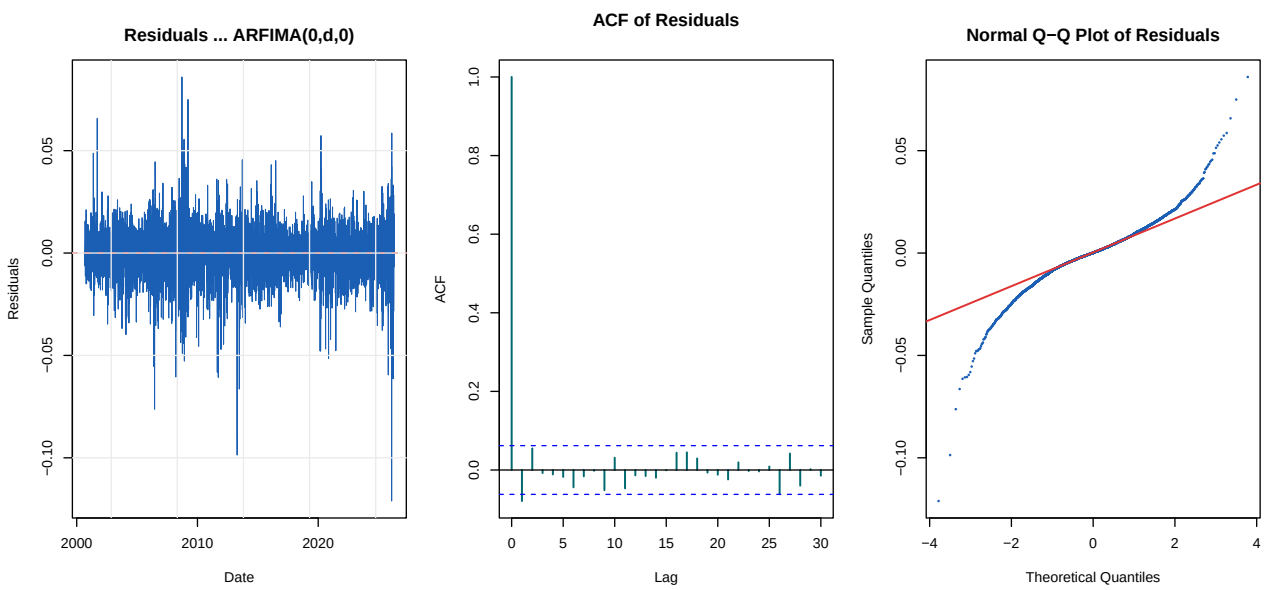

Figure 5: Diagnostic plots for the preferred ARFIMA model. Residuals over time (left) fluctuate around zero with no structure. The ACF (centre) shows all autocorrelations within the 95% confidence bands. The Q-Q plot (right) confirms approximate normality.

The residuals are visually indistinguishable from the original return series — a direct consequence of the near-zero d estimate, meaning the model applies almost no transformation to the data. The ACF shows all autocorrelations within the 95% confidence bands, confirming that no linear serial dependence remains. The Q-Q plot indicates approximate normality in the centre of the distribution, with the mild tail deviations expected for financial return data. Together, the diagnostics confirm model adequacy and validate the white noise conclusion.

5.3 Parametric vs. Nonparametric Spectrum

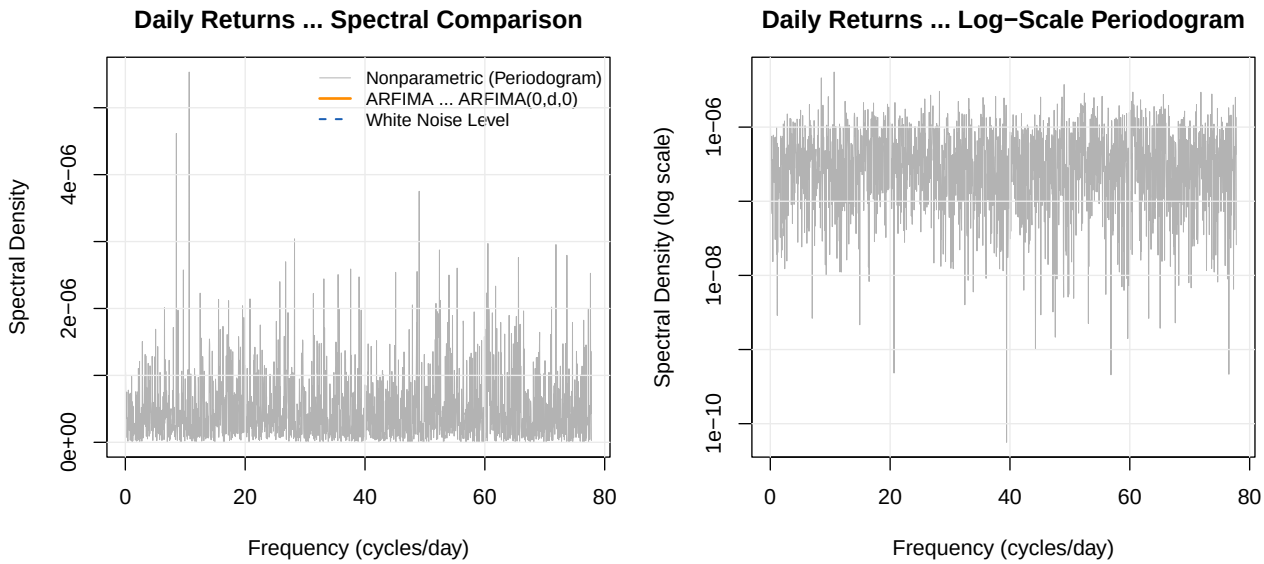

Figure 6: Comparison of the raw periodogram (grey) and the ARFIMA-implied parametric spectrum (orange) for daily gold returns (left), alongside the levels periodogram on a log scale (right). The near-flat ARFIMA spectrum coincides with the white noise reference level, confirming no detectable long-run structure in returns.

The parametric ARFIMA spectrum (orange) is essentially flat across all frequencies, coinciding almost exactly with the theoretical white noise level (blue dashed line). On both the linear and log scales, the ARFIMA spectrum provides no additional shape beyond what a flat white noise reference already describes. This is the final and most direct confirmation of the analysis: unlike Croatian GDP, where the ARFIMA spectrum carved out a clearly declining shape from white noise, gold returns give the long-memory model nothing to explain.

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6 Conclusion

The spectral and long-memory analysis of gold daily returns delivers a remarkably consistent message across all methods and frequencies.

Daily log returns are indistinguishable from white noise by the periodogram test, the cumulative spectral test, the ARFIMA likelihood approach, and the residual ACF. The estimated fractional differencing parameter is essentially zero, no seasonal patterns are detected in monthly data, and the parametric spectrum is flat. In plain terms: **past gold prices contain no linear information about future gold prices.**

This finding aligns directly with the Efficient Market Hypothesis in its weak form. Gold is one of the world's most liquid and globally traded markets — any predictable pattern based on past prices would be quickly identified and traded away, eliminating the pattern itself. The result is not a limitation of the analysis; it is the expected outcome for a well-functioning financial market.

One important caveat deserves emphasis: white noise in returns does not mean white noise in risk. The volatility clustering visible in the squared return plots confirms that **the size of price swings is predictable even when the direction is not.** This distinction is central to financial risk management, where models for time-varying volatility — such as GARCH — would be the natural next step beyond the scope of this report.

This analysis was created by Filip Popovic as part of a data analyst portfolio. Fur questions or inqueries ***contact me here: fpopovic01@gmail.com***